Assessment

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1. **Write 10 test cases**

   LINK

2. **Write a difference between Sanity testing and Regression testing**

Aspect	**Sanity Testing**	**Regression Testing**
Purpose	To quickly check if critical functionalities work correctly after minor changes or fixes.	To ensure that recent code changes or enhancements haven't adversely affected existing functionality.
Scope	Narrow focus on specific areas or components.	Broad scope, covering all or significant portions of the application.
Timing	Typically performed before detailed testing.	Typically performed after functional or major changes.
Depth of Testing	Surface-level testing to verify basic functionality.	Comprehensive testing to detect any regression issues across the application.
Regression Detection	Not focused on regression defects.	Specifically detects regression defects
Timing	Typically performed before detailed testing.	Typically performed after functional or major changes.
Focus	Verifies if changes haven't introduced critical issues.	Ensures overall system integrity and performance post changes.
Test Coverage	Limited to specific tests related to recent changes.	Covers both new functionality and ensures existing functionality remains intact.

3. **Annotate Exploratory testing, Happy path Testing, Unit Testing and Functional Testing**

   Exploratory testing: It is a testing approach where testers perform testing based on their knowledge, experience, and intuition also experience, rather than following predefined steps.

   Happy path Testing: Happy-path testing is a type of software testing that uses known input and produces an expected output. Also referred to as golden-path or sunny-day

testing, the happy-path approach is tightly scripted. It verifies that the system functions correctly under normal conditions without errors or exceptions.

Unit Testing: Unit testing is the process where you test the smallest functional unit of code. Software testing helps ensure code quality, and it's an integral part of software development. It ensures each unit behaves as expected and meets its design specifications.

Functional Testing: Functional testing is a type of software testing where the basic functionalities of an application are tested against a predetermined set of specifications. It involves checking both the functional and non-functional aspects of the application to ensure it performs as intended in various scenarios.

4. **Run a Login and Registration Automation of QA Brain with Selenium**

Prerequisites:

- Install Java Development Kit (JDK): Ensure Java is installed on your system.
- Install Eclipse IDE: Download and set up Eclipse IDE from eclipse.org.
- Set up Selenium WebDriver: Download Selenium WebDriver Java bindings from SeleniumHQ and configure in Eclipse.
- Download ChromeDriver: Get ChromeDriver from ChromeDriver download page and ensure it's in your system's PATH.

Steps to Automate Login and Registration:

1. Create a New Java Project in Eclipse:
- Open Eclipse IDE.
- Go to File -> New -> Java Project.
- Enter a project name (e.g., QA_Automation) and click Finish.

2. Add Selenium WebDriver Dependencies:
- Right-click on your project in the Project Explorer.
- Select Build Path -> Configure Build Path.
- Click on the Libraries tab and then Add External JARs.
- Navigate to where you saved the Selenium WebDriver JAR files, select them, and click Open.

3. Create Test Classes for Login and Registration:

Login Test Class (LoginTest.java):

```
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;

public class LoginTest {
```

```
    public static void main(String[] args) {
        // Set path to ChromeDriver
        System.setProperty("webdriver.chrome.driver", "path_to_chromedriver");

        // Initialize WebDriver
        WebDriver driver = new ChromeDriver();

        // Open QA Brain login page
        driver.get("https://qabrain.com/login");

        // Enter username
        driver.findElement(By.id("username")).sendKeys("your_username");

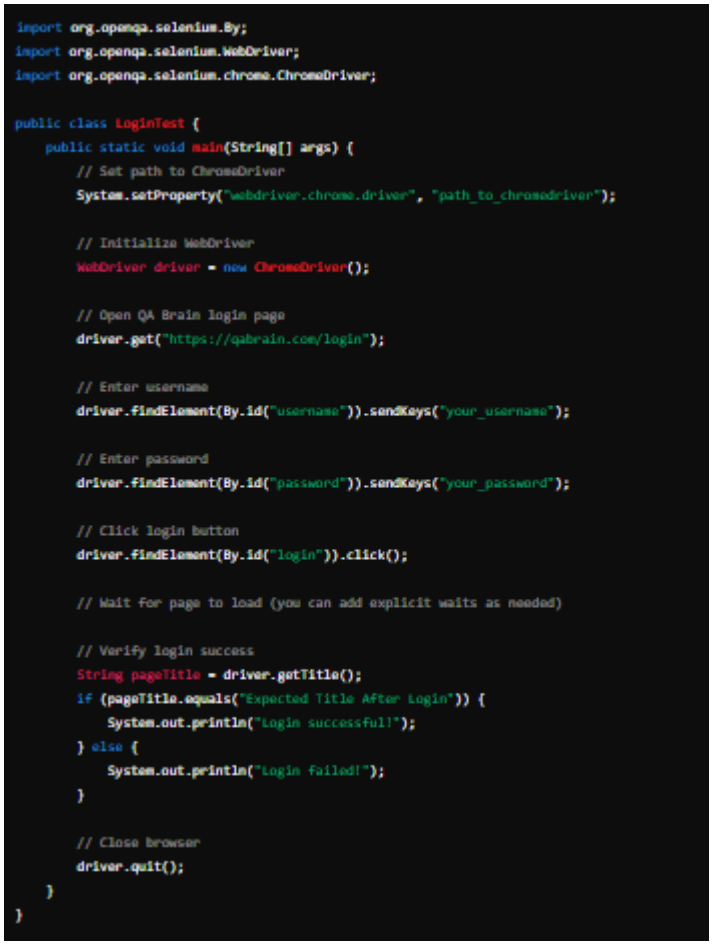
        // Enter password
        driver.findElement(By.id("password")).sendKeys("your_password");

        // Click login button
        driver.findElement(By.id("login")).click();

        // Wait for page to load (you can add explicit waits as needed)

        // Verify login success
        String pageTitle = driver.getTitle();
        if (pageTitle.equals("Expected Title After Login")) {
            System.out.println("Login successful!");
        } else {
            System.out.println("Login failed!");
        }

        // Close browser
        driver.quit();
    }
}
```



```java
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;

public class LoginTest {
    public static void main(String[] args) {
        // Set path to ChromeDriver
        System.setProperty("webdriver.chrome.driver", "path_to_chromedriver");

        // Initialize WebDriver
        WebDriver driver = new ChromeDriver();

        // Open QA Brain login page
        driver.get("https://qabrain.com/login");

        // Enter username
        driver.findElement(By.id("username")).sendKeys("your_username");

        // Enter password
        driver.findElement(By.id("password")).sendKeys("your_password");

        // Click login button
        driver.findElement(By.id("login")).click();

        // Wait for page to load (you can add explicit waits as needed)

        // Verify login success
        String pageTitle = driver.getTitle();
        if (pageTitle.equals("Expected Title After Login")) {
            System.out.println("Login successful!");
        } else {
            System.out.println("Login failed!");
        }

        // Close browser
        driver.quit();
    }
}
```

Registration Test Class (RegistrationTest.java):

```
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;

public class RegistrationTest {
    public static void main(String[] args) {
        // Set path to ChromeDriver
        System.setProperty("webdriver.chrome.driver", "path_to_chromedriver");

        // Initialize WebDriver
        WebDriver driver = new ChromeDriver();

        // Open QA Brain registration page
        driver.get("https://qabrain.com/register");

        // Enter registration details (replace with actual registration fields)
        driver.findElement(By.id("username")).sendKeys("new_username");
        driver.findElement(By.id("email")).sendKeys("test@example.com");
        driver.findElement(By.id("password")).sendKeys("password");
        driver.findElement(By.id("confirmPassword")).sendKeys("password");

        // Click register button
        driver.findElement(By.id("register")).click();
```

```
        // Wait for registration process (you can add explicit waits as needed)

        // Verify registration success (check for success message, redirection, etc.)
        String currentUrl = driver.getCurrentUrl();
        if (currentUrl.equals("Expected URL After Registration")) {
            System.out.println("Registration successful!");
        } else {
            System.out.println("Registration failed!");
        }

        // Close browser
        driver.quit();
    }
}
```

```java
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;

public class RegistrationTest {
    public static void main(String[] args) {
        // Set path to ChromeDriver
        System.setProperty("webdriver.chrome.driver", "path_to_chromedriver");

        // Initialize WebDriver
        WebDriver driver = new ChromeDriver();

        // Open QA Brain registration page
        driver.get("https://qabrain.com/register");

        // Enter registration details (replace with actual registration fields)
        driver.findElement(By.id("username")).sendKeys("new_username");
        driver.findElement(By.id("email")).sendKeys("test@example.com");
        driver.findElement(By.id("password")).sendKeys("password");
        driver.findElement(By.id("confirmPassword")).sendKeys("password");

        // Click register button
        driver.findElement(By.id("register")).click();

        // Wait for registration process (you can add explicit waits as needed)

        // Verify registration success (check for success message, redirection, etc.)
        String currentUrl = driver.getCurrentUrl();
        if (currentUrl.equals("Expected URL After Registration")) {
            System.out.println("Registration successful!");
        } else {
            System.out.println("Registration failed!");
        }

        // Close browser
        driver.quit();
    }
}
```

4. Running the Tests:

- Right-click on each test class (LoginTest.java and RegistrationTest.java).
- Select Run As -> Java Application.
- Eclipse will execute the test in Chrome browser controlled by WebDriver.